General Chemistry I

Chemistry: The central science

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Lecture 13: Chemical Reactions

Chemical Reactions

1. Evidence of a chemical reaction
2. Aqueous solution and solubility
3. Precipitation reactions
4. Acid-base reactions
5. Oxidation-reduction reaction
6. Classifying chemical reactions
7. Working with solution
8. Quantitative Analysis

1. Evidence of a chemical reaction



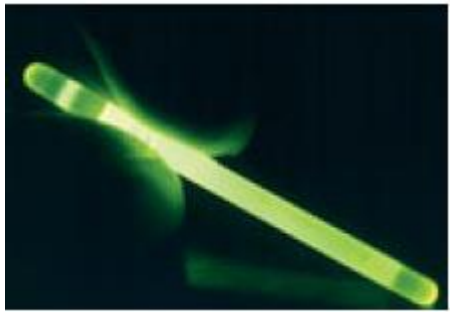
Color change



Gas formation



Solid formation

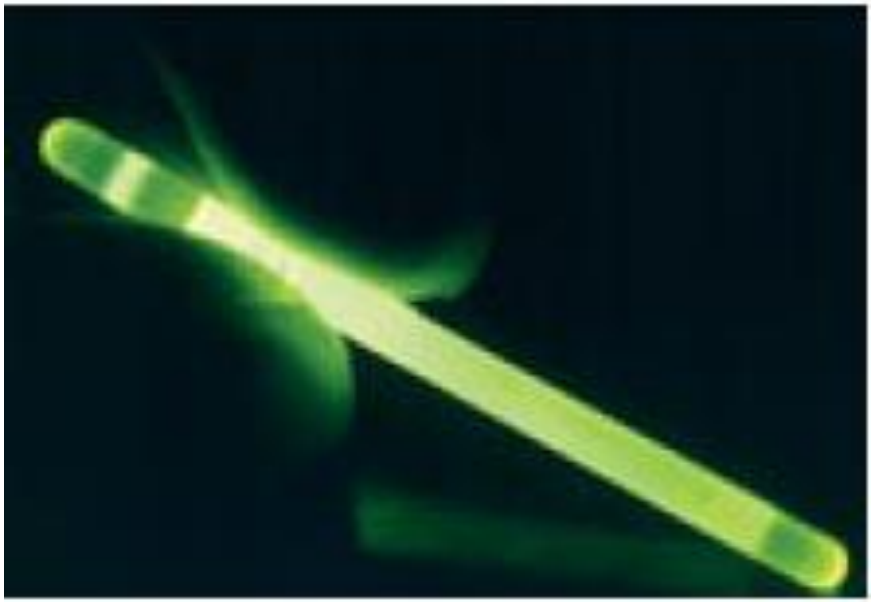


Emission of light

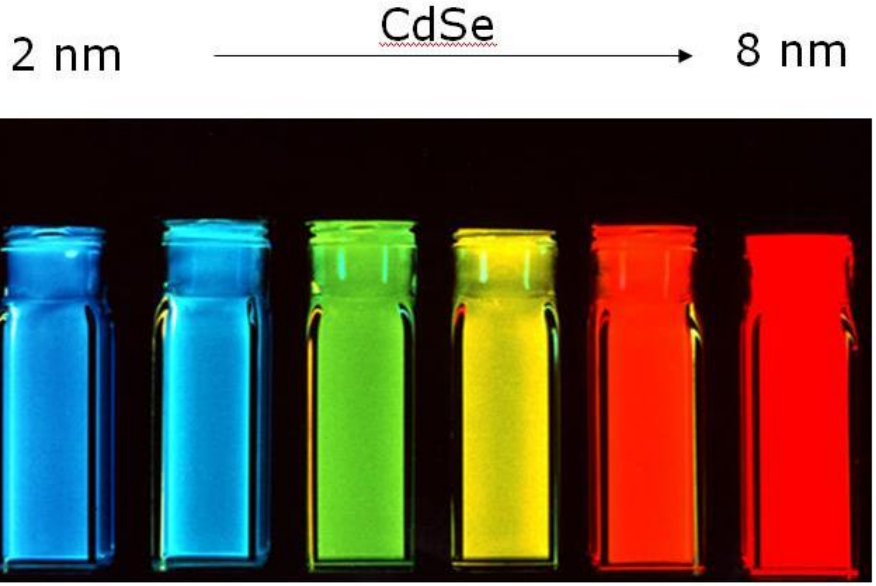


Emission/absorption of heat

Emission of light



A chemical or physical process?



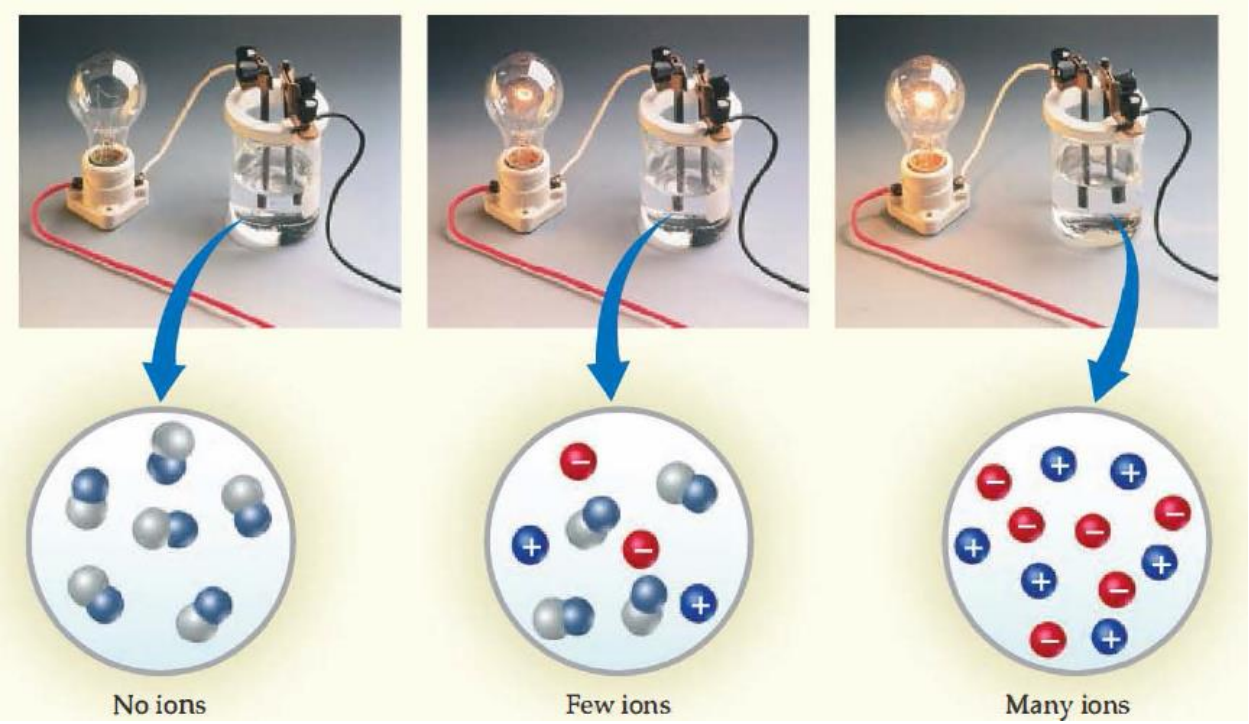
2. Aqueous solution and solubility

- ***A solution*** is a homogeneous mixture of two or more substances
  - ***Solute*****:** the component that is dissolved
  - ***Solvent*****:** the component that does the dissolving

  Solvent is usually the substance present in the greatest quantity
- ***An aqueous solution:*** a solution in which water is the solvent

- Electrolytic properties
  - **Electrolyte:** substance whose aqueous solutions contain ions

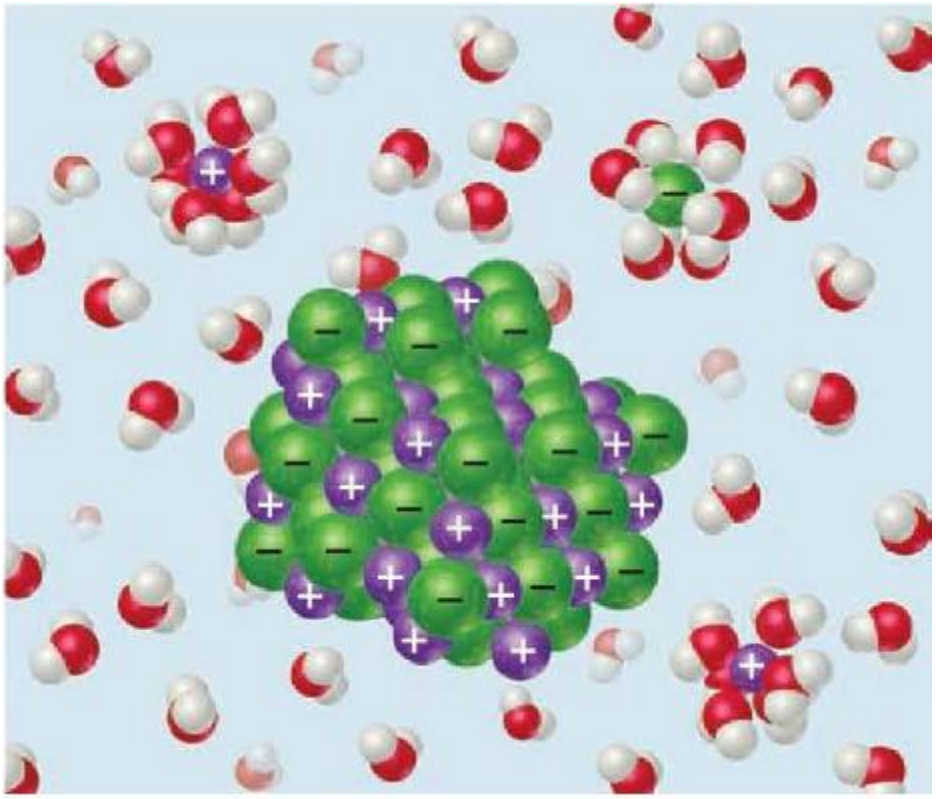
    Electrolyte solution conducts electricity

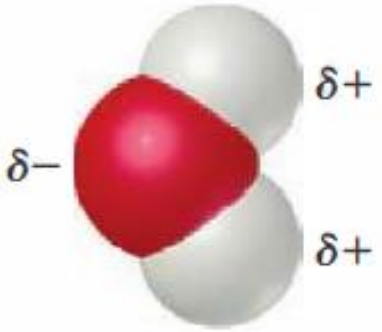


  - **Nonelectrolyte:** substance that does not form ions in solution

$$C_{12}H_{22}O_{11}(s) \xrightarrow{H_2O} C_{12}H_{22}O_{11}(aq)$$

Dissolution in water: NaCl - an ionic compound

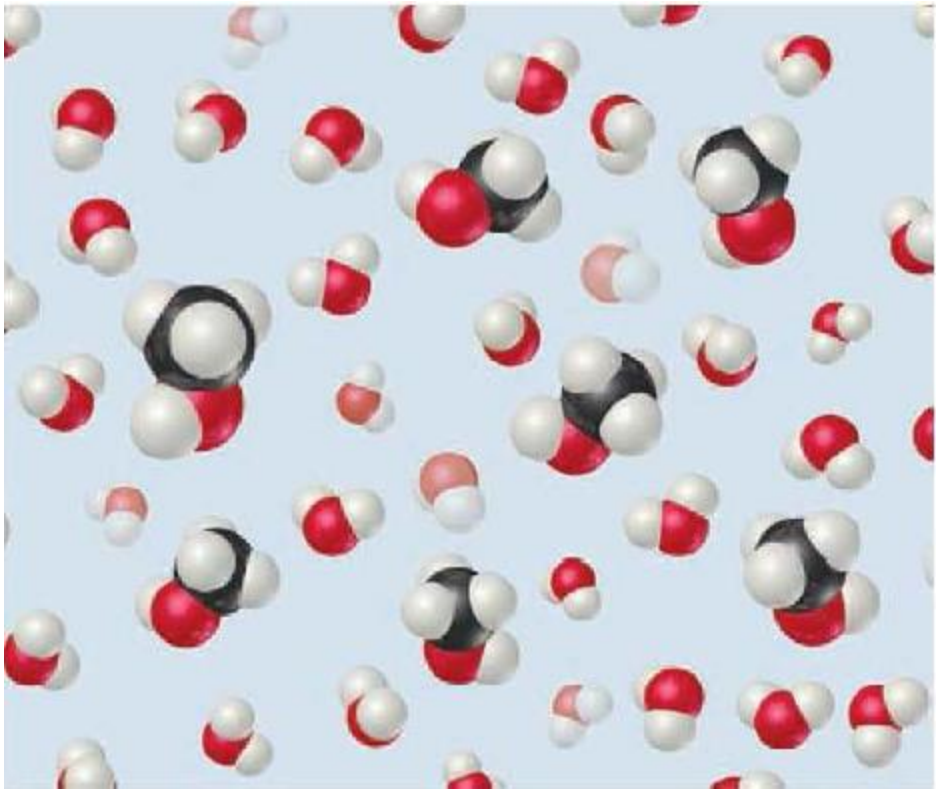




As an ionic compound dissolves, the ions are solvated
i.e surrounded by water molecule and unable to recombine

$$NaCl(s) \xrightarrow{H_2O} Na^+(aq) + Cl^-(aq)$$

Dissolution in water: methanol (a molecular compound)



A molecular compound dissolves without forming ions

- Strong *vs.* weak electrolytes
  - ***Strong Electrolytes***: ~ 100% dissolve in solution as ions

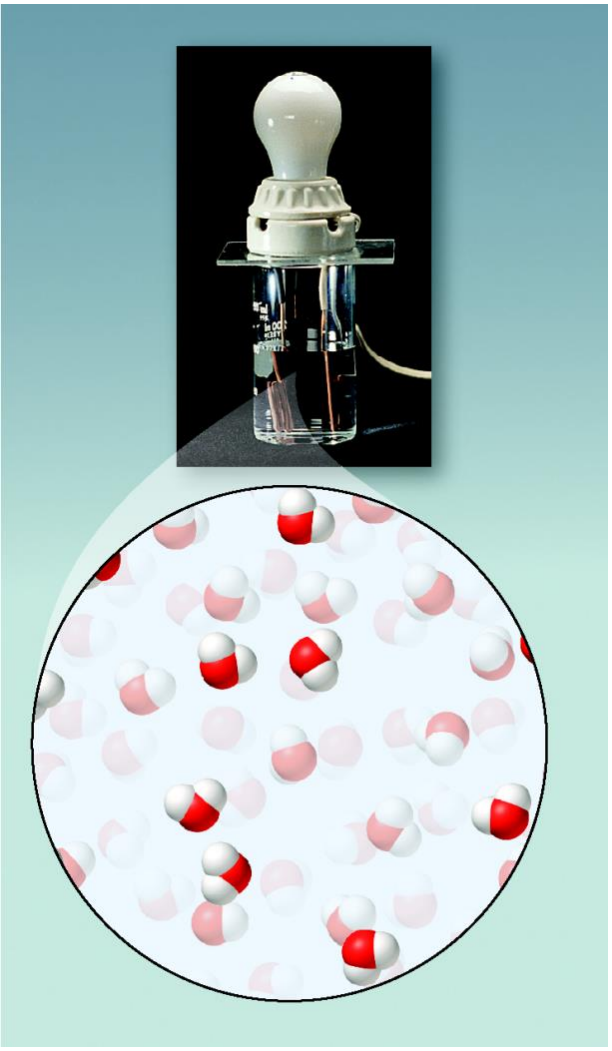
    Example: All water soluble ionic compounds, strong acids and strong bases

$$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$$

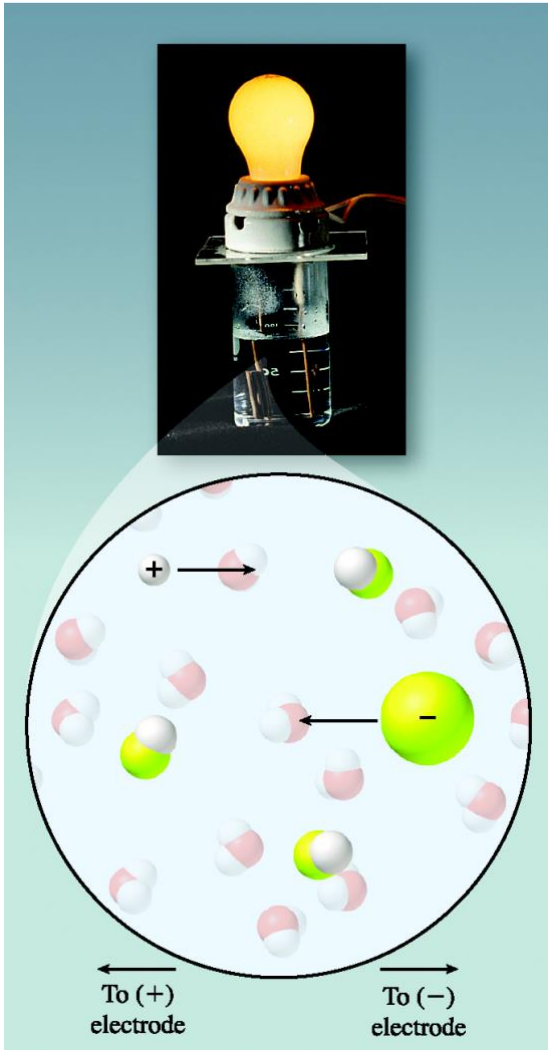
  - ***Weak electrolytes:*** Partially ionized in solution, exist mostly as the molecular form in solution

    Example: Weak acids and weak bases

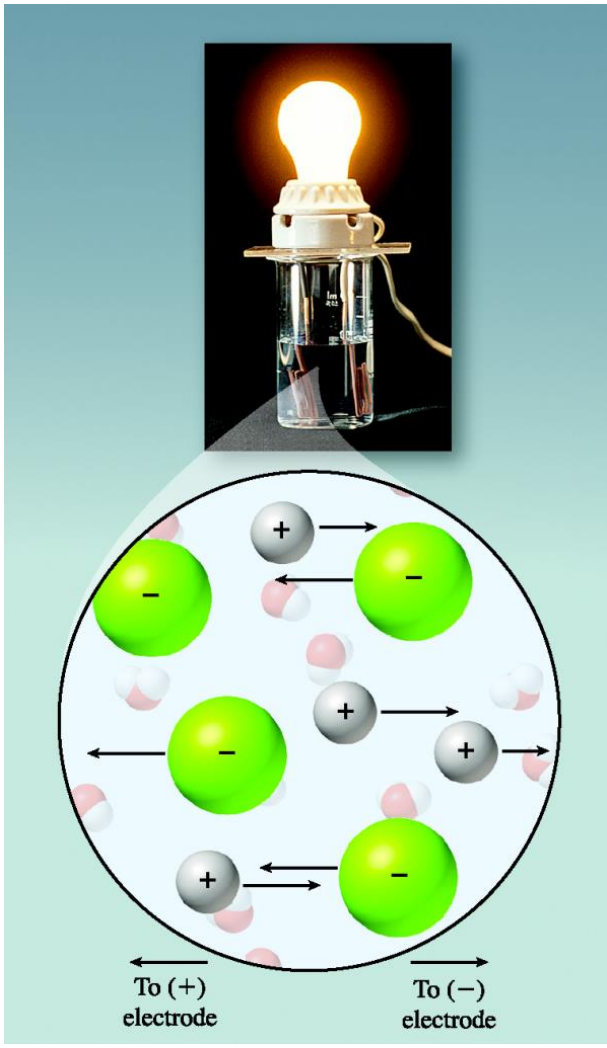
$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$



nonelectrolyte



weak electrolyte



strong electrolyte

Rearrange these compounds follow the order above: NaOH, H_2CO_3, CH_3OH?

- **Solubility***:* ability to dissolve in water
  - Readily dissolve in water: soluble
  - Not dissolve in water: insoluble

Solubility rules for ionic compounds

Rule	Applies to	Statement	Exceptions
1	Li^+, Na^+, K^+, NH_4^+	Group IA and ammonium compounds are soluble.	—
2	$C_2H_3O_2^-$, NO_3^-	Acetates and nitrates are soluble.	—
3	Cl^-, Br^-, I^-	Most chlorides, bromides, and iodides are soluble.	AgCl, Hg_2Cl_2, $PbCl_2$, AgBr, $HgBr_2$, Hg_2Br_2, $PbBr_2$, AgI, HgI_2, Hg_2I_2, PbI_2
4	SO_4^{2-}	Most sulfates are soluble.	$CaSO_4$, $SrSO_4$, $BaSO_4$, Ag_2SO_4, Hg_2SO_4, $PbSO_4$
5	CO_3^{2-}	Most carbonates are insoluble.	Group IA carbonates, $(NH_4)_2CO_3$
6	PO_4^{3-}	Most phosphates are insoluble.	Group IA phosphates, $(NH_4)_3PO_4$
7	S^{2-}	Most sulfides are insoluble.	Group IA sulfides, $(NH_4)_2S$
8	OH^-	Most hydroxides are insoluble.	Group IA hydroxides, $Ca(OH)_2$, $Sr(OH)_2$, $Ba(OH)_2$

Solubility and electrolyte concepts

- *Soluble substances* are either electrolytes or nonelectrolytes
- *Most* ***soluble ionic*** substances are strong electrolytes
- ***Soluble molecular*** substances are nonelectrolytes or weak electrolytes
- A ***few molecular*** substances (such as HCl) are strong electrolytes.

Classifying chemical reactions

1. **Acid – base reaction**
2. **Oxidation-reduction reaction** : involve the transfer of electrons between reactants

Precipitation Reactions

- **A precipitation reaction:** when aqueous solutions of ionic compounds are poured together and a solid (precipitate) forms

  Ex : $MgCl_2$ + $AgNO_3$ → ???



$MgCl_2(aq)$ + 2 $AgNO_3$ (aq) ⟶ 2AgCl (s) + $Mg(NO_3)_2$ (aq)

Insoluble: Any substance with a solubility less than 0.01 mol/L

To predict whether a precipitate forms:

1. Note the ions present in the reactants
2. Consider the possible combinations of the cations and anions
3. Use solubility table to determine if any of these combinations is insoluble

$$Mg(NO_3)_2(aq) + 2\ NaOH(aq) \longrightarrow$$

Solubility rules

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5	CO_3^{2-}	Most carbonates are insoluble.	Group IA carbonates, $(NH_4)_2CO_3$
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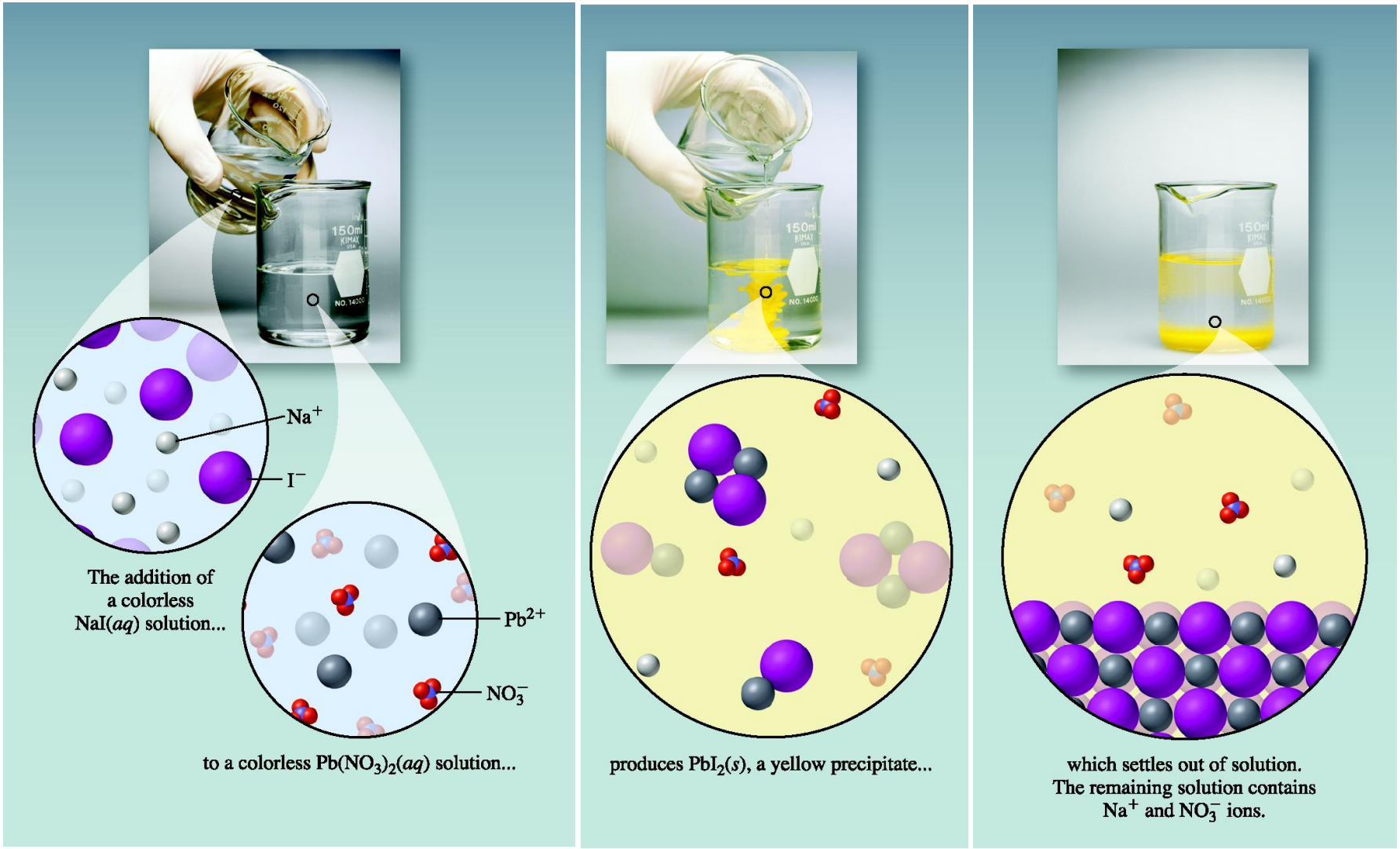
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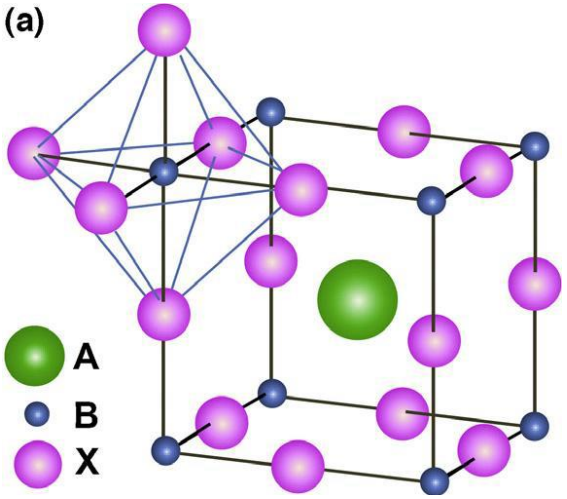
Mixing Solutions of $Pb(NO_3)_2$ and NaCl

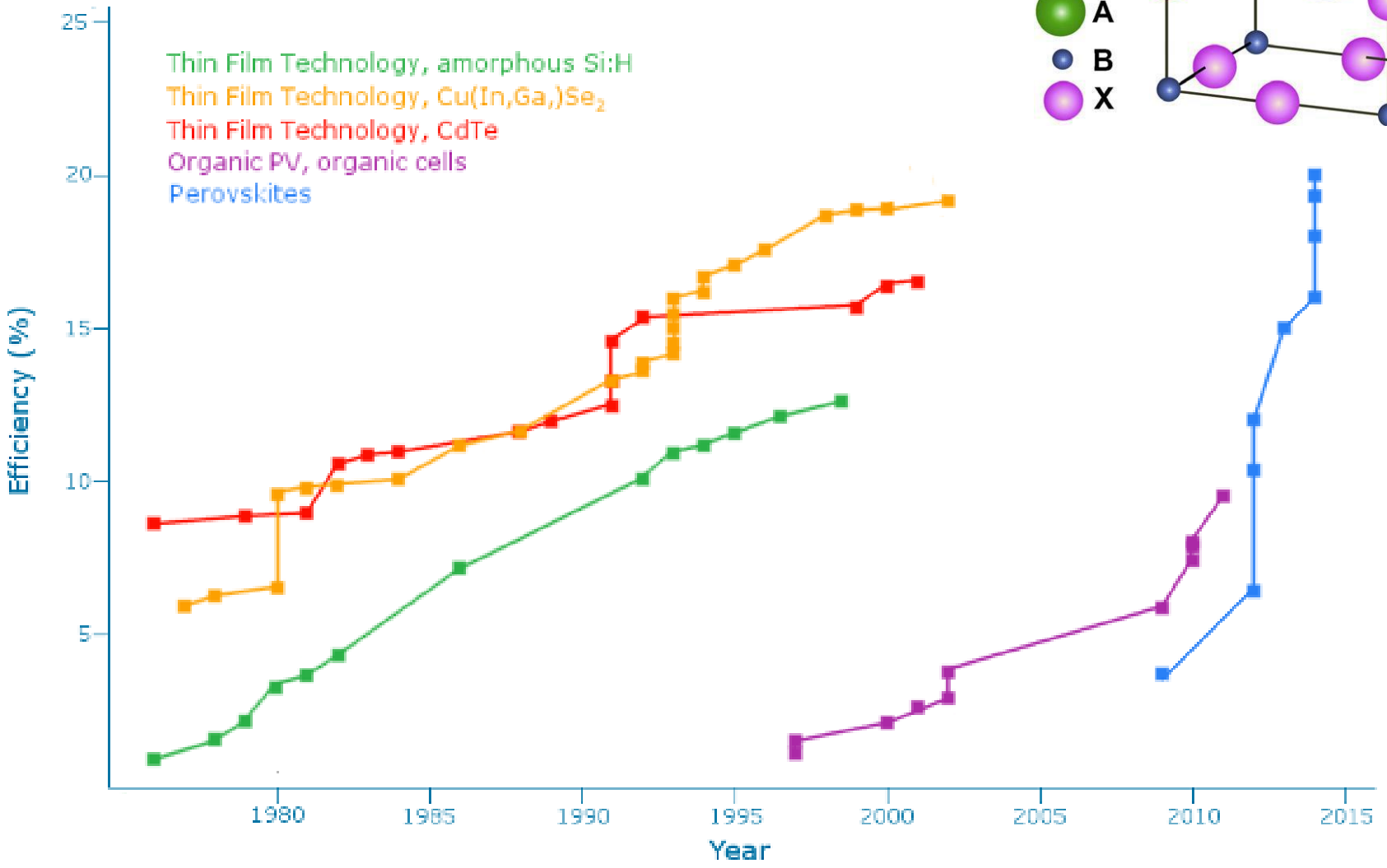


$$Pb(NO_3)_2(aq) + 2NaI(aq) \rightarrow 2NaNO_3(aq) + PbI_2(s)$$

Perovskite emerging solar cell material:

$PbI_2 + CH_3NH_3I \rightarrow CH_3NH_3PbI_3$





Molecular and ionic equations

- ***Molecular equation*:** showing the complete chemical formulas of the reactants and products

$$Na_2SO_4(aq) + Ba(OH)_2(aq) \longrightarrow 2NaOH(aq) + BaSO_4(s)$$

- ***Complete Ionic equation*:** shows all strong electrolytes as ions and all other substances (non-electrolytes, weak electrolytes, gases) by their chemical formulas

$$2Na^+(aq) + SO_4^{2-}(aq) + Ba^{2+}(aq) + 2OH^-(aq) \longrightarrow 2Na^+(aq) + 2OH^-(aq) + BaSO_4(s)$$

- ***Net Ionic equation*:** shows only the reacting species in the chemical equation

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$$

- Steps in writing a net ionic equation

  Step 1: Write the balanced molecular equation.

  - Predict products by exchanging cations and anions in reactants.

  Step 2: Separate strong electrolytes into ions, use solubility rule to predict insoluble product

  Step 3: Cancel spectator ions.

  Step 4: Use the remaining species to write the net ionic equation.

Aqueous solutions of silver nitrate and sodium sulfate are mixed. Write the net ionic reaction.

Step 1:

$2AgNO_3(aq)+Na_2SO_4(aq) \rightarrow 2NaNO_3(?)+Ag_2SO_4(?)$

Step 2: Use solubility table; all nitrates are soluble but silver sulfate is insoluble

$$2Ag^{+}(aq) + 2NO_3^{-}(aq) + 2Na^{+}(aq) + SO_4^{2-}(aq)$$

$$\rightarrow 2Na^{+}(aq) + 2NO_3^{-}(aq) + Ag_2SO_4(s)$$

Step 3: Cancel spectators

$$2Ag^+(aq) + 2NO_3^-(aq) + 2Na^+(aq) + SO_4^{2-}(aq)$$

$$\rightarrow 2Na^+(aq) + 2NO_3^-(aq) + Ag_2SO_4(s)$$

Step 4: Write the net ionic reaction

$$2Ag^+(aq) + SO_4^{2-}(aq) \rightarrow Ag_2SO_4(s)$$

4. Acid – Base reaction

- Definitions of acids and bases
  - ***Arrhenius acid*** - produces H^+ in solution
  - **Arrhenius *base*** - produces OH^- in solution
  - More inclusive definitions:
    - ***Brønsted acid*** - proton donor
    - ***Brønsted base*** - proton acceptor
- **A salt**: a compound that contains a cation other than H^+ and anion other than hydroxide ion, OH^-, or oxide ion, O^{2-}

Some strong acids and their anions

- Common strong acids
  - HCl
  - HBr
  - HI
  - HNO_3
  - $HClO_4$
  - $HClO_3$
  - H_2SO_4
- Anions of these acids
  - Cl^-
  - Br^-
  - I^-
  - NO_3^-
  - ClO_4^- :
  - ClO_3^-:
  - HSO_4^-
  - SO_4^-

Some common weak acids and their anions

- Common weak acids
  - HF
  - CH_3COOH
  - HCN
  - HNO_2
  - H_2CO_3
  - H_2SO_3
  - H_3PO_4
- Anions of these acids
  - F^-
  - CH_3COO^-
  - CN^-
  - NO_2^-
  - HCO_3^-
  - SO_3^{2-}
  - —

Common strong bases

- Group IA
  - LiOH
  - NaOH
  - KOH
  - RbOH
  - CsOH
- Group IIA
  - $Ca(OH)_2$
  - $Sr(OH)_2$
  - $Ba(OH)_2$

Insoluble bases & weak bases

- Insoluble bases: $Cu(OH)_2$, $Zn(OH)_2$, $Fe(OH)_2$ and $Fe(OH)_3$
- Weak bases: NH_3 and closely related N-containing compounds, the amines,

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

Electrolytic Behavior of Common Soluble Ionic and Molecular Compounds

	Strong electrolyte	Weak electrolyte	Non electrolyte
Ionic	All	None	None
Molecular	Strong acids	Weak acids Weak bases	All other compounds

Neutralization reactions

Acid + Base $\rightarrow$ Salt + Water

a reaction of an acid and a base that results in an ionic compound and possibly water

$$2HCl(aq) + Ca(OH)_2(aq) \rightarrow CaCl_2(aq) + 2\,H_2O(l)$$

Acid base salt

$$HCN\,(aq) + KOH\,(aq) \rightarrow KCN\,(aq) + H_2O\,(l)$$

Acid base salt



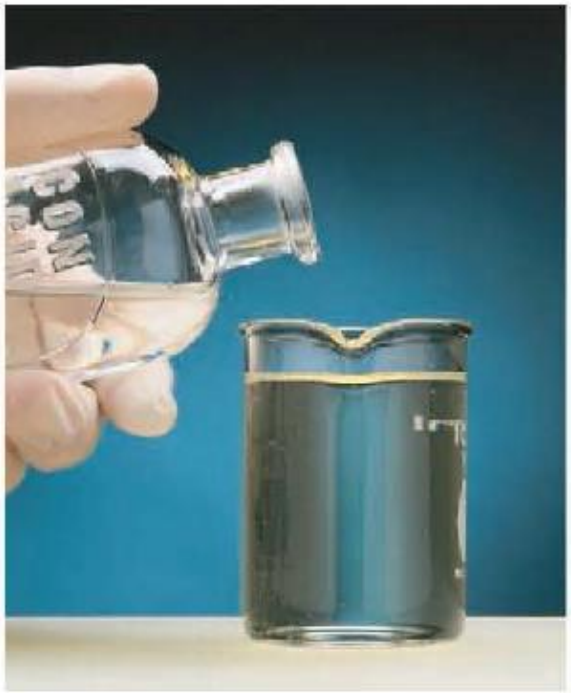
(a)

Milk of Magnesia = Magnesium hydroxide in water



(b)

Magnesium hydroxide dissolves upon the addition of hydrochloric acid



(c)

The final clear solution contains soluble $MgCl_2$ (aq)

Molecular equation:

$$Mg(OH)_2(s) + 2\,HCl(aq) \longrightarrow MgCl_2(aq) + 2\,H_2O(l)$$

Net ionic equation:

$$Mg(OH)_2(s) + 2\,H^+(aq) \longrightarrow Mg^{2+}(aq) + 2\,H_2O(l)$$

Acid-Base Reactions with Gas Formation

- Many bases besides OH^- react with H^+ to form molecular compound

Molecular equation: $2\,HCl(aq) + Na_2S(aq) \longrightarrow H_2S(g) + 2\,NaCl(aq)$

Net ionic equation: $2\,H^+(aq) + S^{2-}(aq) \longrightarrow H_2S(g)$

Molecular equation:

$HCl(aq) + NaHCO_3(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$

Net ionic equation: $H^+(aq) + HCO_3^-(aq) \longrightarrow H_2O(l) + CO_2(g)$

5. Oxidation-reduction reaction (Redox reaction)

Reactions involving a transfer of electrons from one species to another







The molecular equation for this reaction is:

$$Fe(s) + CuSO_4(aq) \longrightarrow FeSO_4(aq) + Cu(s)$$

The net ionic equation is:

$$Fe(s) + Cu^{2+}(aq) \longrightarrow Fe^{2+}(aq) + Cu(s)$$

Oxidation numbers

- ***Oxidation numbers:*** hypothetical charge assigned to the atom, assuming that the electrons are completely held by one atom or the other

>>> *a way to keep track of electron transfer*

Rules for assigning oxidation numbers

1. Elements (uncombined) are 0.

   Al, N_2, He, Zn, Ag, Br_2, O_2, O_3

2. Monatomic ion: oxidation number = ion charge

   $K^+ = +1$

   $S^{2-} = -2$

3. Nonmetals usually have negative oxidation numbers

   O =-2 usually

   F =-1 usually

4. Sum of the oxidation number = overall charge

   H_3O^+= +1, H = +1,O = -2

   Overall = 3(+1)+(-2) = +1

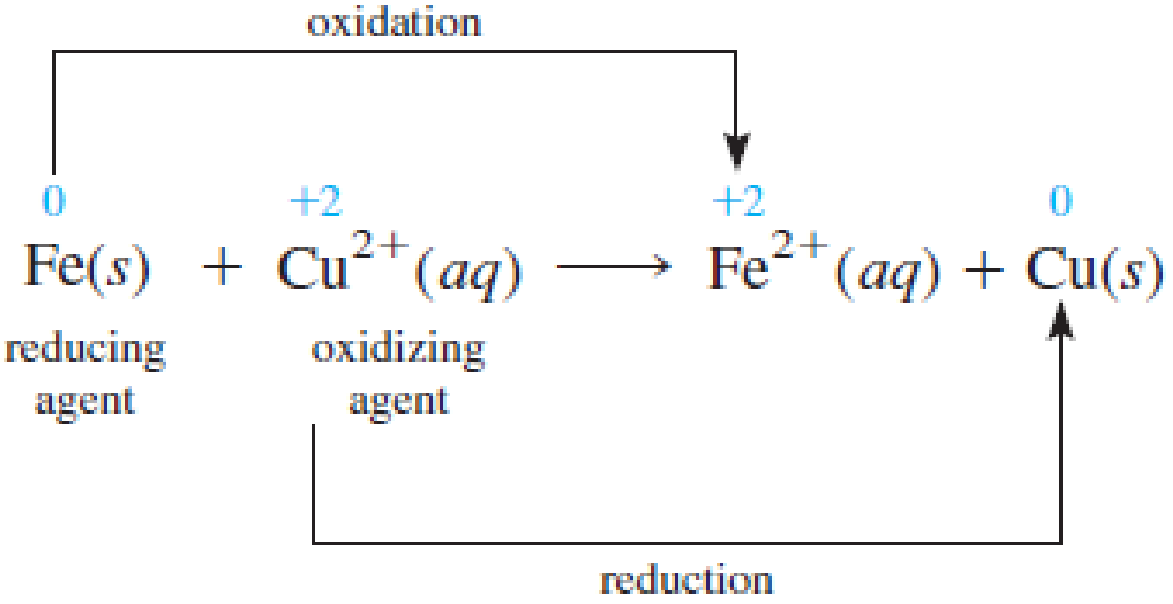
Assign oxidation numbers for all elements in each species

SO_2

$MgBr_2$

ClO_2^-

- ***Oxidation*** happens when a substance loses electrons, the substance is ***oxidized***
- ***Reduction*** happens when a substance gains electrons, the substance is ***reduced***
- Oxidation and reduction occur simultaneously

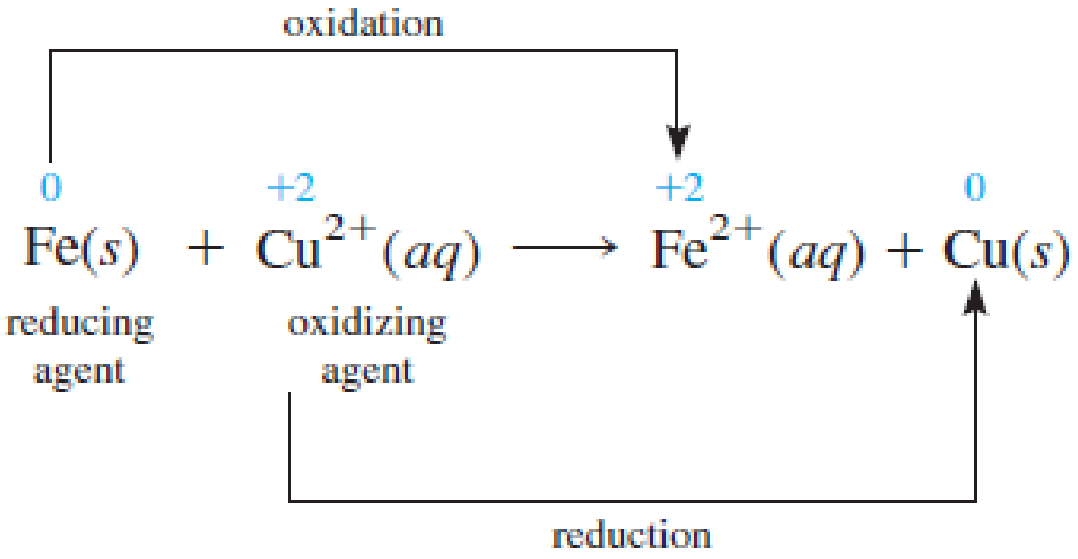


$$\overset{0}{\mathrm{Fe}(s)} \longrightarrow \overset{+2}{\mathrm{Fe}^{2+}}(aq) + 2\mathrm{e}^- \qquad \text{(electrons lost by Fe)}$$

$$\overset{+2}{\mathrm{Cu}^{2+}}(aq) + 2\mathrm{e}^- \longrightarrow \overset{0}{\mathrm{Cu}}(s) \qquad \text{(electrons gained by } \mathrm{Cu}^{2+}\text{)}$$

Oxidizing and reducing agents

	Oxidizing	Reducing
1	Oxidize other substances	Reduce other substances
2	Contain atoms that are reduced	Contain atoms that are oxidized
3	Gain electrons	Lose electrons



- Displacement reactions
  - A common reaction: active metal replaces (displaces) a metal ion from a solution

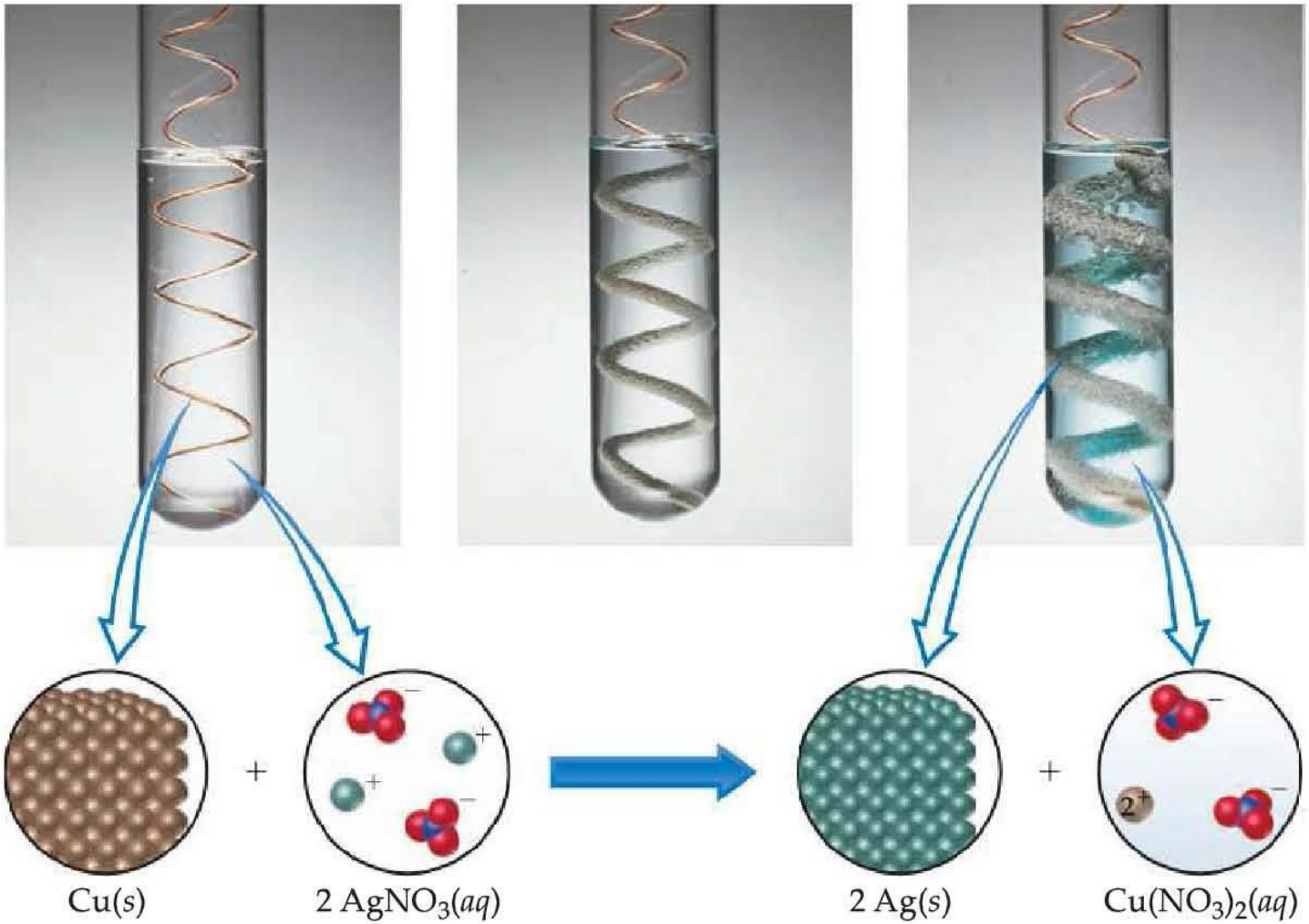
$$Mg(s) + CuCl_2(aq) \rightarrow Cu(s) + MgCl_2(aq)$$

  - The **activity series** of metals is useful in order to predict the outcome of the reaction.

Activity Series

Metal	Oxidation Reaction
Lithium	$Li(s) \longrightarrow Li^+(aq) + e^-$
Potassium	$K(s) \longrightarrow K^+(aq) + e^-$
Barium	$Ba(s) \longrightarrow Ba^{2+}(aq) + 2e^-$
Calcium	$Ca(s) \longrightarrow Ca^{2+}(aq) + 2e^-$
Sodium	$Na(s) \longrightarrow Na^+(aq) + e^-$
Magnesium	$Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$
Aluminum	$Al(s) \longrightarrow Al^{3+}(aq) + 3e^-$
Manganese	$Mn(s) \longrightarrow Mn^{2+}(aq) + 2e^-$
Zinc	$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$
Chromium	$Cr(s) \longrightarrow Cr^{3+}(aq) + 3e^-$
Iron	$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-$
Cobalt	$Co(s) \longrightarrow Co^{2+}(aq) + 2e^-$
Nickel	$Ni(s) \longrightarrow Ni^{2+}(aq) + 2e^-$
Tin	$Sn(s) \longrightarrow Sn^{2+}(aq) + 2e^-$
Lead	$Pb(s) \longrightarrow Pb^{2+}(aq) + 2e^-$
Hydrogen	$H_2(g) \longrightarrow 2\,H^+(aq) + 2e^-$
Copper	$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$
Silver	$Ag(s) \longrightarrow Ag^+(aq) + e^-$
Mercury	$Hg(l) \longrightarrow Hg^{2+}(aq) + 2e^-$
Platinum	$Pt(s) \longrightarrow Pt^{2+}(aq) + 2e^-$
Gold	$Au(s) \longrightarrow Au^{3+}(aq) + 3e^-$





$$Cu(s) + 2\,Ag^{+}(aq) \longrightarrow Cu^{2+}(aq) + 2\,Ag(s)$$

Notice that this net ionic reaction is balanced based on:

- Types and number of atoms
- Charges of ions

Example

- Given two substances: $FeCl_2$ and Mg, will they react with each other?

6. Classifying chemical reaction

- We can also classify chemical reaction based on what atoms do.
- 4 categories:

Type of Reaction	Generic Equation
synthesis or combination	$A + B \longrightarrow AB$
decomposition	$AB \longrightarrow A + B$
displacement	$A + BC \longrightarrow AC + B$
double-displacement	$AB + CD \longrightarrow AD + CB$

- **Synthesis or combination reaction:** simpler substances combine to form more complex substances

$$2\,Na(s) + Cl_2(g) \longrightarrow 2\,NaCl(s)$$

$$CaO(s) + CO_2(g) \longrightarrow CaCO_3(s)$$

$$A + B \longrightarrow AB$$

- **Decomposition reaction:** a complex substance decomposes to form simpler substances

$$2\,H_2O(l) \xrightarrow{\text{electrical current}} 2\,H_2(g) + O_2(g)$$

$$CaCO_3(s) \xrightarrow{\text{heat}} CaO(s) + CO_2(g)$$

$$AB \longrightarrow A + B$$

- **Displacement reaction:** one element displaces another in a compound

$$Zn(s) + CuCl_2(aq) \longrightarrow ZnCl_2(aq) + Cu(s)$$

$$Mg(s) + 2\,HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

$$2\,Na(s) + 2\,H_2O(l) \longrightarrow 2\,NaOH(aq) + H_2(g)$$

$$A + BC \longrightarrow AC + B$$

- **Double displacement reaction:** two elements or groups of elements in two different compounds exchange places to form two new compounds

$$AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(aq)$$

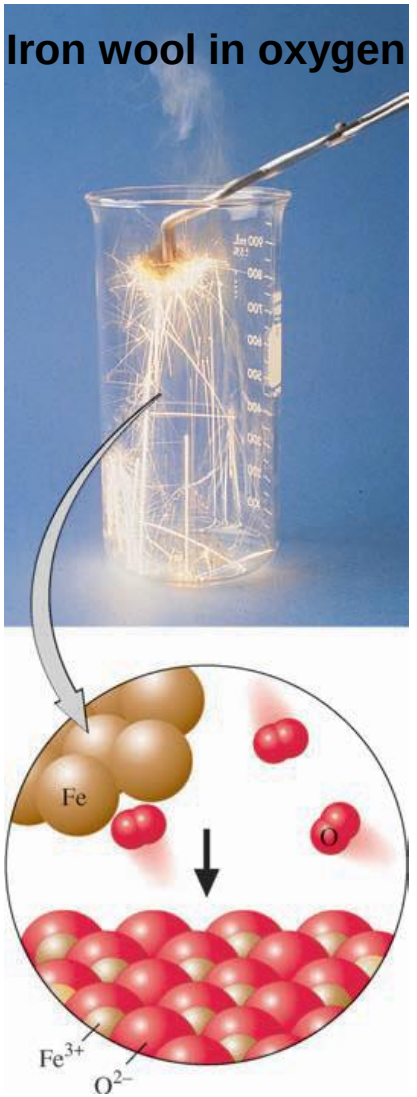
$$HCl(aq) + NaOH(aq) \longrightarrow H_2O(l) + NaCl(aq)$$

$$2\,HCl(aq) + Na_2CO_3(aq) \longrightarrow H_2CO_3(aq) + 2\,NaCl(aq)$$

$$AB + CD \longrightarrow AD + CB$$

Combustion reactions

- A **combustion reaction** is *a reaction in with oxygen, usually with the rapid release of heat to produce a flame*
- The products include one or more oxides
- Oxygen changes oxidation number from 0 to 2
- Ex : $4Fe_{(s)} + 3O_{2(g)} \rightarrow 2Fe_2O_{3(s)}$



Combustion reaction is a special type of combination reaction

7. Working with Solutions

7.1 Molar Concentration

7.2 Diluting Solutions

Solutions

- Solution = homogenous mixture
- Solute = what is being dissolved
- Solvent = what is doing the dissolving
- “Soluble” means will dissolve

Concentration

- The quantity of solute in a given volume of solution which is usually 1 liter (dm^3)

$$Concentration\ (g.dm^{-3}) = \frac{mass\ of\ solute\ (g)}{Volume\ of\ solution\ (dm^3)}$$

Molarity (Molar concentration)

- The number of moles of solute that are present in 1 liter (dm^3) of solution

$$Molarity\ (mol.dm^{-3}) = \frac{number\ of\ moles\ of\ solute\ (mol)}{volume\ of\ solution\ (dm^3)}$$

Relationship between Molarity and concentration

Molarity ($mol.dm^{-3}$)

× molar mass

÷ molar mass

Concentration ($g.dm^{-3}$)

Preparation of a Solution by Dilution Method



Preparing a solution by diluting a concentrated one



Concentrated solution
(High Molarity)

Add water



diluted solution
(Low Molarity)

moles of solute before dilution = moles of solute after dilution

$$M_1V_1 = M_2V_2$$

M_1 : Molarity of the solution before water is added
V_1 : Volume of the solution before water is added
M_2 : Molarity of the solution after water is added
V_2 : Volume of the solution after water is added

For the next experiment the class will need 250. mL of 0.10 *M* $CuCl_2$. There is a bottle of 2.0 *M* $CuCl_2$.

Describe how to prepare this solution.

How much of the 2.0 *M* solution do we need?

Concentrated: 2.0 *M* use ? mL (V_2)

Dilute: 250. mL of 0.10 *M*

$$M_1V_1 = M_2V_2$$

$$(2.0\ M)\ (V_1) = (0.10\ M)\ (250.\text{mL})$$

$$V_1 = 12.5\ \text{mL}$$

12.5 mL of the concentrated solution are needed; add enough distilled water to prepare 250. mL of the solution.

- Solution Stoichiometry
  - Soluble ionic compounds dissociate completely in solution.
  - Using mole ratios we can calculate the concentration of all species in solution.

    NaCl dissociates into Na^+ and Cl^-

    Na_2SO_4 dissociates into $2Na^+$ and SO_4^{2-}

    $AlCl_3$ dissociates into Al^{3+} and $3Cl^-$

Find the concentration of all species in a 0.25 *M* solution of $MgCl_2$

$$MgCl_2 \rightarrow Mg^{2+} + 2Cl^-$$

Given: $MgCl_2 = 0.25\ M$

$[Mg^{2+}] = 0.25\ M$ (1:1 ratio)

$[Cl^-] = 0.50\ M$ (1:2 ratio)

8. Quantitative Analysis

Quantitative Analysis *: determination of the amount of a substance or species present in a material*

8.1 Gravimetric Analysis

8.2 Volumetric Analysis

- Gravimetric Analysis
  - One form: isolation of a precipitate
  - Typical steps:
    - Determine mass of unknown solid
    - Dissolve unknown in water
    - Combine with excess amount of known substance to form a precipitate (excess drives reaction to completion)
    - Filter, dry and weigh the precipitate
    - Use formula and mass of ppt to find % of ion in unknown solid



A solution of potassium chromate (yellow) is poured down a stirring rod into a solution containing an unknown amount of barium ion, Ba^{2+}



barium chromate, $BaCrO_4$ solution is filtered by pouring it into a crucible containing a porous glass partition. Afterward, the crucible is heated to dry the barium chromate. By weighing the crucible before and afterward, you can determine the mass of precipitate.

Volumetric Analysis

Volumetric analysis :

- An important method for determining the amount of a particular substance is based on measuring the volume of reactant solution
- *A method of analysis based on titration*

Titration : *a procedure for determining the amount of substance* A *by adding a carefully measured volume of a solution with known concentration of* B *until the reaction of* A *and* B *is just complete.*

Titration of an unknown amount of HCl with NaOH



The flask contains HCl and a few drops of phenolphthalein indicator; the buret contains 0.207 *M NaOH (the buret reading* is 44.97 mL).



NaOH was added to the solution in the flask until a persistent faint pink color was reached, marking the endpoint of the titration (the buret reading is 49.44 mL) The amount of HCl can be determined from the volume of NaOH used (4.47 mL);



The addition of several drops of NaOH solution beyond the endpoint gives a deep pink color.

Key points

- Electrolytes (strong, weak, and non)
- Precipitation reactions
  - Solubility rules
- Molecular, ionic, and net ionic reactions
- Acid-base neutralization reactions
- Oxidation-reduction reactions: oxidation number, activity series
- Molarity
- Solution stoichiometry
  - Gravimetric analysis
  - Volumetric analysis

General Chemistry I

Chemistry: The central science

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Section 12: Chemical Reactions